



Day 2 Notes



React Basics – Simple Explanation

★ Today = first steps into React. No project code yet — just a practice folder.

1 What is a Component?

A component is just a small, reusable piece of a webpage. Think of it like a LEGO block.

You build small blocks (like a "ProductCard") and use them again and again with different info.

- A component is just a JavaScript function
- It returns what should show on the screen

2 What is JSX?

JSX looks like HTML, but it is written inside JavaScript. React turns it into a real webpage.

```
function App() {  
  return <h1>Hello!</h1>  
}
```

Curly braces { } let you put a JavaScript value inside JSX. Example: {count} shows the count's value.

3 What is State?

State = data that a component remembers, and can change over time (like a counter number).

```
const [count, setCount] = useState(0)
```

- count = the current value (starts at 0)
- setCount = the function used to change it

When you call setCount, React automatically updates the screen. You never touch the page manually.

4 What are Props?

Props = information you pass INTO a component, so the same component can show different things.

```
function ProductCard(props) {  
  return (  
    <div>  
      <h3>{props.name}</h3>  
      <p>${props.price}</p>  
    )  
}
```

```
</div>
)
}
```

Then used like this, with different info each time:

```
<ProductCard name="Blue T-Shirt" price="19.99" />
<ProductCard name="Red Hoodie" price="39.99" />
```

Same component, two different results. That's the whole point of props!

5 Showing a List with `.map()`

Instead of writing a `ProductCard` by hand for every product, we can loop over a list of data.

```
const products = [
  { name: 'Blue T-Shirt', price: 19.99 },
  { name: 'Red Hoodie', price: 39.99 },
  { name: 'Black Cap', price: 12.5 },
]

{products.map((product) => (
  <ProductCard name={product.name} price={product.price} key={product.name} />
))}
```

This turns a data list into a list of cards automatically. This is exactly how a real product catalog works!

Note: React always wants a unique key on each item in a list, so it can keep track of which is which.



Big takeaway:

Component → holds a piece of UI. State → data that can change. Props → data passed in from outside. `.map()` → turns a list of data into a list of components. These four ideas are the base of almost everything in React.

— made while building: *AI Virtual Try-On E-commerce App* —